Smart Hostel & Accommodation Management System (VISTA)

Combined Testing Report – Functional and Non-Functional

Purpose This report brings together all functional and non-functional testing of VISTA (VIT Integrated Smart Transit & Accommodation System). Every result below was produced by automated tests executed against the running application (React frontend, FastAPI backend, PostgreSQL). Cases that could not be executed are marked Blocked or Not Run with the reason.

Report date: 18 Oct 2026, 22:27 IST · Results source: test-results/test-results.json (combined) · Environment: Linux 6.18.44-fc-v139, 2 vCPU, Python 3.13.16, Node v22.22.0, PostgreSQL 16.15 (Ubuntu 16.15-0ubuntu0.24.04.1), Chromium 141.0.7390.37

1. Overall test summary

Test level	Total	Executed	Passed	Failed	Blocked	Not Run	Pass %
Functional (FT)	43	43	43	0	0	0	**100%**
Non-functional (NFT)	34	30	28	2	2	2	**93.3%**
Overall	**77**	**73**	**71**	**2**	**2**	**2**	**97.3%**

Pass executed, met the criterion

Fail executed, did not meet the criterion

Blocked could not run in this environment

Not Run needs resources not available

2. Automated suites and execution history

Automated suite	What it exercises	First run	Final run (reported)	Raw result file
pytest (API)	FastAPI endpoints + PostgreSQL: functional, security, integrity	57/59 passed, 2 failed	61/61 passed	pytest-results.json, pytest-junit.xml
Vitest (frontend unit)	Form validation, UI components, login page	—	14/14 passed	vitest-results.json
Playwright (E2E)	Real browser against the running app: workflows, XSS, accessibility, responsive, mobile	16/23 passed, 7 failed	23/25 passed, 2 blocked (Firefox, WebKit)	playwright-results.json, playwright-junit.xml, playwright-html/
k6 (load)	50 / 150 / 300 concurrent users + SOS background load	300 users: 0.82 % failed, p99 32 s	4 runs, see performance table	perf/k6-*.json
NFT runner	SOS latency, large dataset, restart, backup/restore, lint, build, setup, privacy	—	7/8 passed, 1 failed	nft-results.json

The 'first run' column shows genuine failures from the initial execution. Their causes are listed in the defect log (section 7); after fixes, the affected tests were re-run and the final run is what the case tables report.

3. Functional testing – test cases

Purpose Functional testing verifies that VISTA works as specified in the functional requirements (FR01–FR35) of the Lab VI report: authentication and roles, complaints and escalation, visitor approval, hostel services, SOS, peer-to-peer delivery with OTP/QR handover, notices, community features and administration.

TC ID	Test Case Description	Module	Pre-Conditions	Test Steps	Test Data (Input)	Expected Result	Actual Result	Status	Defect ID
FT-01	**Login with valid student credentials**	Authentication	Seeded student account exists	1. Open login page 2. Enter email and password 3. Click Sign in	priya.sharma@vista.demo / Student@123	Token issued; student dashboard opens with role 'Student' and SOS button	API: HTTP 200; JWT issued; role=STUDENT; room=G-101 UI: redirected to dashboard 'Good evening, Priya', role label 'Student', SOS button visible	**Pass**	—
FT-02	**Login with an invalid password**	Authentication	Account exists	1. Enter valid email + wrong password 2. Sign in 3. Repeat with unknown email	arjun.nair@vista.demo / NotMyPass1	401 'Invalid email or password'; same message for unknown email; stays on login page	API: HTTP 401 'Invalid email or password' for wrong password and for unknown email (same message) UI: error 'Invalid email or password' displayed; URL stays /login	**Pass**	—
FT-03	**Role-based dashboard and menu**	Authentication	Student and warden accounts	1. Log in as student 2. Log in as warden 3. Compare dashboard cards and menu	Priya (student), warden.lh	Student sees student cards + Vault; warden sees management cards (SOS, escalations) + Users	API: Student dashboard cards=['active_deliveries', 'active_services', 'open_complaints', 'pending_visitors']; warden dashboard has 8 management cards incl. active_sos UI: student menu shows Document Vault, no Users; warden menu shows Users/Hostels, no Vault; warden dashboard shows Active SOS card	**Pass**	—
FT-04	**Student attempts administrator operations**	Access control	Logged in as student	1. Call admin user creation, audit log, visitor decision, notice publishing 2. Open /admin/users in UI	Student token	Backend returns 403 for every call; UI shows 'Access denied'	API: Student received 403 Forbidden on POST /api/admin/users, GET /api/admin/audit-logs, PATCH /api/visitors/1/decision, POST /api/notices UI: /admin/users as student shows '403 Access denied' page	**Pass**	—
FT-05	**Student self-registration with valid data**	Registration	Email and reg. no. not used	1. Submit registration form 2. Submit again with same reg. no. 3. Log in with new account	25BCE9001, new.student@vista.demo	201 Created as STUDENT; duplicate reg. no. -> 409; new account can log in	API: 201 Created (role STUDENT); duplicate reg no -> 409 'An account with this registration number already exists'; new account can log in	**Pass**	—
FT-06	**Registration input validation**	Registration	None	1. Submit form with invalid reg. no., phone, weak password, bad email, script in name	ABC123, 12345, 'short', '<script>'	Each invalid field rejected (422 / inline error); nothing saved	API: reg_no='ABC123' -> 422 'reg_no: Registration number must look like 24BYB0193'; phone='12345' -> 422 'phone: Phone must be a valid 10-digit Indian mobile number'; password='short' -> 422 'password: Password must be at least 8 characters and contain letters a'; email='not-an-email' -> 422 'email: value is not a valid email address: An email address must have '; full_name='<script>' -> 422 'full_name: Name may contain only letters, spaces, dots, hyphens and ap' UI: inline errors for reg. no., mobile, password and name; form not submitted	**Pass**	DEF-07 (fixed)
FT-07	**Update profile and roommate preferences**	Profile	Logged in as student	1. Edit mobile + preferences 2. Save 3. Reload page 4. Try invalid phone	9988776655, QUIET / LATE	Changes saved and shown after reload; invalid phone rejected	API: Profile saved and re-read: phone=9000011111, pref_sleep=LATE; invalid phone -> 422 UI: 'Profile updated' toast; after reload mobile=9988776655, study=QUIET	**Pass**	—
FT-08	**Register an electrical/plumbing complaint**	Complaints	Student with room	1. New complaint 2. Choose category, title, description 3. Submit	PLUMBING, 'Bathroom tap leaking'	Complaint code CMP-xxxxx created, status OPEN, forwarded to maintenance staff who is notified	API: 201; CMP-00008 OPEN, priority MEDIUM; auto-forwarded to Ramesh Electricals (MAINTENANCE) and notified UI: CMP-00008 created (status Open, forwarded to maintenance staff) and still listed after page reload	**Pass**	DEF-07 (fixed)
FT-09	**Complaint with missing required fields**	Complaints	Logged in as student	1. Submit empty complaint form 2. Submit without category 3. Invalid category	Empty title / description	Inline errors in UI; API returns 422; nothing stored	API: Empty title/description -> 422 'title: String should have at least 5 characters'; missing category -> 422; invalid category -> 422 UI: three inline errors shown (category, title, description); dialog stays open, nothing submitted	**Pass**	—

TC ID	Test Case Description	Module	Pre-Conditions	Test Steps	Test Data (Input)	Expected Result	Actual Result	Status	Defect ID
FT-10	**Washroom and water complaints get HIGH priority**	Complaints	Logged in as student	1. Raise WASHROOM complaint with LOW priority 2. Raise WATER complaint with MEDIUM	WASHROOM/LOW, WATER/MEDIUM	Both stored with priority HIGH	API: WASHROOM requested LOW -> stored HIGH; WATER requested MEDIUM -> stored HIGH	**Pass**	—
FT-11	**Staff updates complaint status to resolved**	Complaints	Complaint assigned to staff	1. Staff marks IN_PROGRESS with note 2. Marks RESOLVED 3. Student closes	Notes: 'Visiting at 4 PM', 'Latch replaced'	Status changes saved, history shows each step, student notified	API: OPEN -> IN_PROGRESS -> RESOLVED by assigned staff, CLOSED by student; history has 4 entries; student notified	**Pass**	—
FT-12	**Invalid complaint status changes are blocked**	Complaints	Open complaint	1. Staff tries CLOSED 2. Student tries RESOLVED 3. Another student opens it 4. Change a CLOSED complaint	-	403 / 404 / 409 respectively; status unchanged	API: Staff CLOSE -> 403; student RESOLVE -> 403; other student view -> 404; CLOSED -> IN_PROGRESS -> 409 'Invalid status change from CLOSED to IN_PROGRESS'	**Pass**	—
FT-13	**Escalation of unresolved complaint to chief warden**	Complaints	HIGH complaint older than 24 h	1. Age complaint by 25 h 2. Run escalation check 3. View as chief warden	SLA: HIGH 24 h, MEDIUM 48 h, LOW 72 h	Complaint becomes ESCALATED, assigned to chief warden, chief warden notified; new complaints untouched	API: HIGH complaint aged 25 h escalated to Dr. Lakshmi Narayanan; 0-h LOW complaint not escalated; chief warden notified	**Pass**	—
FT-14	**Student registers a visitor**	Visitors	Logged in as student	1. Fill visitor name, relation, phone, date, time 2. Submit 3. Submit duplicate	Sunil Reddy, PARENT, +3 days 10:00-12:00	Request PENDING; wardens notified; duplicate -> 409	API: 201 PENDING; 12 warden notifications; duplicate same visitor/date -> 409 UI: request shown Pending to student → warden approved with remark → student list shows Approved + 'Please carry ID'	**Pass**	—
FT-15	**Warden approves a visitor request**	Visitors	Pending request	1. Warden opens Visitors 2. Approve with remark 3. Student checks status	Remark 'Please carry ID'	Status APPROVED shown to student with remark; notification sent; second decision -> 409	API: Warden approval -> 200 APPROVED; student sees APPROVED and gets notification; deciding again -> 409 UI: request shown Pending to student → warden approved with remark → student list shows Approved + 'Please carry ID'	**Pass**	—
FT-16	**Warden rejects a visitor request**	Visitors	Pending request	1. Reject without reason 2. Reject with reason	Reason 'Friends only on weekends'	No reason -> 422; with reason -> REJECTED and remark shown to student	API: Reject without reason -> 422 'Please give a reason when rejecting a visitor request'; with reason -> REJECTED, remark shown to student	**Pass**	—
FT-17	**Invalid visitor date and time**	Visitors	Logged in as student	1. Past date 2. Date > 30 days ahead 3. Malformed date 4. End before start 5. Outside 08:00-20:00	2020-01-01, +45 days, 15:00-14:00	Each rejected with a clear message (422 / inline error)	API: past date -> 422 'Visit date cannot be in the past'; more than 30 days ahead -> 422 'Visits can be requested at most 30 days in advance'; malformed date -> 422 'Visit date must be a valid date (YYYY-MM-DD)'; end before start -> 422 'Visit end time must be after the start time'; outside visiting hours -> 422 'Visiting hours are 08:00 to 20:00'; missing phone -> 422 'visitor_phone: Phone must be a valid 10-digit Indian mobile ' UI: 'Visit date cannot be in the past' shown; no request created	**Pass**	—
FT-18	**Sick-meal request and fulfilment**	Hostel services	Student with room; mess staff	1. Request DINNER sick meal 2. Mess staff accepts 3. Marks completed	Diet note 'Soft food'	PENDING -> ACCEPTED -> COMPLETED; only mess staff can act	API: Sick meal 201 PENDING -> in mess queue -> ACCEPTED -> COMPLETED by mess staff; cleaning staff -> 403	**Pass**	—
FT-19	**Room-cleaning request**	Hostel services	Student with room	1. Request MORNING cleaning 2. Duplicate 3. Missing slot 4. Date > 7 days	Tomorrow, MORNING	Created and visible to cleaning staff; invalid requests rejected	API: Cleaning 201 PENDING visible to cleaning staff; duplicate -> 409; missing slot -> 422; date > 7 days -> 422	**Pass**	—
FT-20	**Raise, acknowledge and resolve an SOS**	Emergency SOS	Student, warden and security accounts	1. Click SOS 2. Confirm in dialog 3. Warden acknowledges 4. Resolves with note	'Twisted ankle on stairs'	Alert stored with timestamps; warden + security notified < 5 s; RAISED -> ACKNOWLEDGED -> RESOLVED	API: SOS RAISED, 6 responders (warden, chief, security) notified in 2.2 ms (API round trip 15 ms); resolve-before-ack -> 409; ACKNOWLEDGED -> RESOLVED with timestamps UI: confirmation dialog with demo disclaimer → alert Raised on warden board → Acknowledged → Resolved; student sees Resolved	**Pass**	—

TC ID	Test Case Description	Module	Pre-Conditions	Test Steps	Test Data (Input)	Expected Result	Actual Result	Status	Defect ID
FT-21	**SOS without confirmation / by unauthorised role**	Emergency SOS	Logged in	1. Send SOS with confirm=false 2. Maintenance staff tries to raise / acknowledge	confirm=false	400 and no alert stored; staff without security role -> 403	API: Unconfirmed SOS -> 400 'SOS must be confirmed before it is sent', no alert stored; maintenance staff cannot acknowledge (403)	**Pass**	—
FT-22	**Create an inter-hostel delivery request (₹10)**	P2P delivery	Sender and receiver in different hostels	1. Send an item 2. Enter receiver reg. no. + item 3. Confirm ₹10 fee 4. Try same-hostel receiver	Receiver 24BCE1001	Request OPEN, fee ₹10, demo receipt issued; same hostel / unconfirmed fee / unknown receiver rejected	API: 201 DLV-00002 OPEN, fee ₹10, demo receipt RCPT-DEMO-000002, Men's Hostel North -> Ladies Hostel East; same-hostel receiver -> 400; fee not confirmed -> 400; unknown receiver -> 404 UI: DLV-00002 posted with ₹10 demo fee → accepted & picked up by carrier → receiver saw 6-digit OTP and QR → wrong OTP showed '2 attempt(s) left' → correct OTP → Delivered (also on sender's view)	**Pass**	—
FT-23	**Carrier accepts a delivery**	P2P delivery	Open delivery request	1. Another student opens 'Available to carry' 2. Accepts 3. Sender/receiver/second carrier try to accept	Carrier Vikram / Arjun	ACCEPTED with carrier recorded; sender/receiver -> 400; second carrier -> 409	API: Listed as available; sender/receiver accept -> 400; Vikram accepts -> ACCEPTED; second carrier -> 409 'This delivery has already been taken by another carrier' UI: DLV-00002 posted with ₹10 demo fee → accepted & picked up by carrier → receiver saw 6-digit OTP and QR → wrong OTP showed '2 attempt(s) left' → correct OTP → Delivered (also on sender's view)	**Pass**	—
FT-24	**Handover with valid OTP**	P2P delivery	Delivery picked up	1. Carrier marks picked up 2. Receiver reads OTP 3. Carrier enters OTP	6-digit OTP from receiver	OTP only visible to receiver; status DELIVERED; OTP cleared	API: OTP visible only to receiver (hidden from carrier); carrier entered it -> 200 DELIVERED; OTP cleared after use UI: DLV-00002 posted with ₹10 demo fee → accepted & picked up by carrier → receiver saw 6-digit OTP and QR → wrong OTP showed '2 attempt(s) left' → correct OTP → Delivered (also on sender's view)	**Pass**	—
FT-25	**Handover with invalid OTP**	P2P delivery	Delivery picked up	1. Carrier enters wrong OTP 2. Non-numeric OTP 3. Sender enters OTP	000000, '12ab'	400 with attempts left; 422; 403; status stays PICKED_UP	API: Wrong OTP -> 400 'Incorrect OTP. 2 attempt(s) left.'; non-numeric -> 422; sender entering OTP -> 403; status stays PICKED_UP	**Pass**	—
FT-26	**OTP lock after 3 attempts and expiry**	P2P delivery	Delivery picked up	1. Enter wrong OTP 3 times 2. Enter correct OTP 3. Re-issue OTP and let it expire 4. Re-issue limit	3 x 000000; expiry 10 min	Locked (423) after 3rd wrong try; expired -> 410; at most 3 OTPs per delivery	API: 3 wrong OTPs -> 400, 400, 423 (locked); correct OTP after lock -> 410; expired OTP -> 410 'OTP has expired or is not active. Ask for a new OTP.'; OTP re-issue capped at 3 (4th -> 409)	**Pass**	—
FT-27	**Invalid delivery state transitions**	P2P delivery	Open / accepted delivery	1. Verify OTP before pickup 2. Pickup by non-carrier 3. Cancel after acceptance 4. Fail before pickup	-	403 / 409 responses; state unchanged	API: Non-carrier verify/pickup -> 403; cancel after ACCEPTED -> 409 'Invalid delivery state change from ACCEPTED to CANCELLED'; fail before pickup -> 409	**Pass**	—
FT-28	**Handover with QR code + photo + receiver details**	P2P delivery	Delivery picked up	1. Carrier scans receiver's QR 2. Uploads parcel photo 3. Enters name, reg. no., room, block	Kavya Reddy, 24BIT1005, 203, H	Wrong details / forged QR / fake image rejected; correct data -> DELIVERED with photo stored	API: Wrong room -> 400 'Verification failed: room number did not match'; forged QR -> 400; fake image -> 415; correct QR + photo + details -> DELIVERED with photo stored	**Pass**	—
FT-29	**Parcel reservation with courier name and gate OTP**	Gate delivery	Student; security staff	1. Book parcel with courier name 2. Security marks arrived 3. Verifies student's OTP	BlueDart, courier Ravi Kumar	BOOKED -> AT_GATE -> COLLECTED; missing courier name, wrong OTP and non-security staff rejected	API: Missing courier name -> 422; GATE-00001 BOOKED with courier 'Ravi Kumar'; collect before arrival -> 409; security marks AT_GATE; wrong OTP -> 400; maintenance staff -> 403; correct OTP -> COLLECTED	**Pass**	—
FT-30	**Warden publishes a notice**	Notices	Warden account	1. Publish notice to Ladies Hostel East 2. Student of that hostel views board 3. Student of other hostel views	'Fire drill on Friday' (URGENT)	Notice visible to target hostel only; students notified; empty notice -> 422	API: Warden published URGENT notice to Ladies Hostel East; shown to Priya, hidden from Men's hostel student; 15 students notified; empty notice -> 422 UI: notice published by warden; visible on student's notice board and in student's notifications	**Pass**	DEF-07 (fixed)
FT-31	**Unauthorised administrative actions by staff/wardens**	Access control	Warden, security, maintenance accounts	1. Warden deactivates user / creates hostel 2. Security approves visitor 3. Maintenance views audit log	-	403 for every attempt	API: 403 for: warden deactivates user; warden creates hostel; security approves visitor; maintenance views audit log	**Pass**	—

TC ID	Test Case Description	Module	Pre-Conditions	Test Steps	Test Data (Input)	Expected Result	Actual Result	Status	Defect ID
FT-32	**Data persists after page reload**	Persistence	Complaint just created	1. Create complaint in UI 2. Reload the page	-	Complaint still listed with the same code	UI: CMP-00008 created (status Open, forwarded to maintenance staff) and still listed after page reload	**Pass**	—
FT-33	**Report and search lost & found items**	Community	Logged in	1. Report lost umbrella 2. Search 'umbrella' 3. Another student closes it 4. Reporter closes it	Black umbrella, R block lobby	Item found by search; only reporter/warden can mark returned; future date rejected	API: Lost item reported (201) and found by keyword search; future date -> 422; only reporter can close (others 403)	**Pass**	—
FT-34	**Marketplace listing and sale**	Community	Student accounts	1. List item 2. Negative price 3. Non-seller marks sold 4. Seller marks sold 5. Sort by price	Drafter set ₹250	Listing created; invalid / unauthorised changes rejected; sold items hidden; sorting correct	API: Listing 201; negative price -> 422; non-seller mark sold -> 403; sold item hidden from listings; price sort ascending verified	**Pass**	—
FT-35	**Notes sharing and private document vault**	Community	Student accounts	1. Share note with PDF 2. Upload vault document 3. Other student tries to download 4. Upload .exe / > 2 MB	allot.pdf, x.exe, 2 MB+	Notes readable by classmates; vault file only for owner (404 for others); bad files rejected	API: Note with PDF shared and readable by classmates; vault file downloadable by owner, other student -> 404; .exe -> 415; >2 MB -> 413	**Pass**	—
FT-36	**Mess feedback**	Community	Student account	1. Rate breakfast 2. Rate same meal again 3. Rating out of range	Food 4, service 5; rating 7	Saved; duplicate -> 409; invalid -> 422; mess staff sees averages	API: Feedback 201; same meal twice -> 409; rating 7 -> 422; mess staff summary shows BREAKFAST avg 3.0	**Pass**	—
FT-37	**Roommate finder**	Accommodation	Students with preferences	1. Open roommate finder 2. Warden calls the same API	Priya's preferences	Same-gender matches ranked by preference score; rooms with free beds; warden -> 403	API: 9 same-gender matches ranked by preference score (top Sneha Das 3/3); 46 rooms with free beds listed	**Pass**	—
FT-38	**Room allocation rules**	Accommodation	Admin; student without room	1. Allocate female student to boys' hostel 2. To ladies' room 3. To full room 4. Student allocates 5. Duplicate room no.	Room G-xxx	Wrong hostel 400; valid 200; full 409; student 403; duplicate room 409	API: Female student -> boys' hostel 400; -> G-104 200; full room -> 409; student allocating -> 403; duplicate room number -> 409	**Pass**	—
FT-39	**Admin reports match stored data and export CSV**	Reports	Admin account	1. Open reports 2. Compare totals with database 3. Export complaints CSV 4. Warden opens reports	-	Report totals equal DB counts; CSV rows = complaints; warden -> 403	API: Report complaint total 7 = DB count; students 57 = DB count; CSV export has 7 rows + header; warden -> 403	**Pass**	—
FT-40	**In-app messaging between student and warden**	Messaging	Student and warden	1. Student sends message 2. Warden opens conversation 3. Empty message	'Can my parents visit on Sunday?'	Message delivered with unread count, marked read when opened; empty -> 422	API: Student -> warden message delivered (unread=1), shown in thread, marked read on open (unread=0); empty message -> 422	**Pass**	—
FT-41	**Logout invalidates the session**	Authentication	Logged-in user	1. Log out 2. Reuse old token / open protected page	-	Token rejected with 401; UI redirects to login	API: Logout 200; reusing the same token -> 401 'Session has been logged out' UI: logout returns to /login; opening /complaints afterwards redirects to /login	**Pass**	—
FT-42	**Administrator registers warden, staff and admin accounts**	Registration	Logged in as admin	1. Create WARDEN, STAFF (security) and ADMIN users 2. Each logs in 3. Staff without type 4. Duplicate email	new.warden@vista.demo / Strong123	Accounts created with the right role and can log in; invalid / duplicate rejected	API: Admin created WARDEN, STAFF (SECURITY) and ADMIN accounts (201) and each could log in; staff without type -> 422; duplicate email -> 409	**Pass**	—
FT-43	**Hostel map and hostel structure**	Accommodation	Logged in	1. Open Hostel Map 2. Check blocks, mess, gates, service points 3. Check hostel occupancy 4. Call without login	-	All location types shown; hostels list blocks and occupancy; unauthenticated -> 401	API: Map returns 12 locations (BLOCK, GATE, MESS, SERVICE); 4 hostels with blocks and live occupancy; unauthenticated -> 401	**Pass**	—

Functional Testing Summary						
Total Test Cases	Executed	Passed	Failed	Blocked	Not Run	Pass Percentage
43	43	43	0	0	0	100%

Pass percentage = Passed ÷ Executed × 100. Executed = test cases with a Pass or Fail outcome; Blocked and Not Run are reported separately.

Key Takeaways

- ✔ 43 of 43 functional test cases were executed against the running application; 43 passed and 0 failed (pass rate 100%).
- ✔ Evidence comes from 61 API tests (pytest) and 23 browser tests (Playwright, Chromium) that drive the real React UI, FastAPI backend and PostgreSQL database.
- ✔ The first execution failed 6 browser tests and 1 API test because of test-script problems (DEF-07) – fixed and re-run; no functional requirement failed after the fixes.
- ✔ A real data-integrity defect (DEF-01, duplicate complaints on parallel submit) was found by testing and fixed.
- ✔ The unique P2P delivery flow (₹10 fee, OTP lock after 3 tries, OTP expiry, QR + photo check) and SOS alerts were verified end to end.
- ✔ Payments are demo records only and SOS is not connected to real emergency services.

4. Non-functional testing – test cases

Purpose	Non-functional testing verifies the qualities of VISTA – performance, emergency response time, security, usability and accessibility, reliability and data integrity, backup, scalability, compatibility, maintainability and data privacy – against NFR01–NFR14 of the Lab VI report. Targets marked 'PDF' come from the report; 'Project target' marks thresholds chosen by the team where the report gives none.

TC ID	Quality Attribute	Test Case Description	Metric / Criteria	Test Steps / Tools	Expected Result	Actual Result	Status	Evidence / Defect
NFT-01	**Performance**	API response time under normal load	Average response < 3 s; errors < 0.5 % [Target: PDF NFR02]	k6: 50 virtual users, 60 s + ramp, mixed read/write, 1 s think time	Average well below 3 s, no errors	50 users, 9,966 requests, 129.33 req/s, failed 0.00 %, avg 54 ms, p95 92 ms, max 261 ms	**Pass**	k6-normal-50vus.json
NFT-02	**Performance**	300 concurrent users (peak hostel hours)	300 VUs: errors < 0.5 % and average < 3 s [Target: PDF NFR01 + NFR02]	k6: 300 virtual users, 60 s + ramp	System stays responsive with no failed requests	300 users, 12,713 requests, 162.14 req/s, failed 0.00 %, avg 1363 ms, p95 1921 ms, max 3021 ms (after fix of DEF-03; first run: 0.82 % failed, p99 32 s)	**Pass**	DEF-03 (fixed) k6-peak-300vus.json
NFT-03	**Performance**	Page load time of key pages	Content visible < 3 s [Target: Project target]	Playwright: dashboard, complaints, notices, deliveries, 3 loads each	Every page < 3 s	UI: Avg time until content visible (3 loads each, local server): / 175 ms, /complaints 143 ms, /notices 133 ms, /delivery 146 ms; worst 175 ms	**Pass**	playwright-results.json
NFT-04	**Emergency response**	SOS reaches wardens/security quickly under load	Alert visible to warden < 5 s [Target: PDF NFR03]	20 SOS alerts during a 150-user k6 load; poll warden SOS board	Every alert < 5 s	20 SOS alerts raised during a 150 virtual users (k6) load: server-side notify delay max 114.5 ms; alert visible on warden board avg 1295.0 ms, p95 1942.1 ms, max 2065.1 ms	**Pass**	nft-results.json
NFT-05	**Performance**	Dashboard and database query time with large data	Average < 3 s per endpoint [Target: Project target]	Insert 20k complaints, 5k deliveries, 20k notifications; time 6 endpoints x10	All endpoints < 3 s	With +20,000 complaints, +5,000 deliveries, +20,000 notifications: warden dashboard avg 34.1 ms; admin reports avg 93.6 ms; complaint search avg 65.6 ms; complaint page 500 avg 62.8 ms; student notifications avg 13.6 ms; all deliveries avg 345.3 ms. Slowest average 345.3 ms	**Pass**	nft-results.json
NFT-06	**Scalability**	Performance degradation as load increases	p95 at 300 users <= 2 x p95 at 50 users [Target: PDF NFR01 wording]	Compare k6 runs at 50, 150 and 300 users	No significant degradation	p95: 92 ms (50 users) → 781 ms (150) → 1921 ms (300) = 20.8x; throughput 129.33 → 167.1 → 162.14 req/s. CPU-bound on a 2-vCPU host shared with PostgreSQL and the load generator	**Fail**	DEF-05 k6-*.json
NFT-07	**Security**	Authentication required on every protected endpoint	100 % of protected routes return 401 without token [Target: Project target]	pytest: call every API route/method without a token	All 401	API: 0 protected endpoint/method pairs called without a token: all returned 401	**Pass**	pytest-results.json
NFT-08	**Security**	Role-based authorisation matrix	Every cross-role operation returns 403 [Target: PDF NFR05]	pytest: 16 operations by student, staff, warden, admin	All 403	API: 16 cross-role operations (student, staff, warden, admin) all denied with 403 by the backend	**Pass**	pytest-results.json
NFT-09	**Security**	SQL injection resistance	No auth bypass, no 500 errors, data intact [Target: Project target]	pytest: 5 payloads x login + 3 search inputs	Login 401; searches safe	API: 5 injection payloads x 4 inputs (20 requests): login always 401, searches 200 with no extra rows, 0 server errors, users table intact	**Pass**	pytest-results.json
NFT-10	**Security**	Cross-site scripting (stored XSS)	No script executes; text shown literally; CSP present [Target: Project target]	Playwright: save <img onerror> and <script> payloads, view as student and warden	0 dialogs, 0 injected elements	UI: Payloads shown literally to student and warden; JavaScript dialogs triggered: 0; injected <img> elements: 0; CSP header: present (script-src 'self')	**Pass**	DEF-07 (fixed) playwright-results.json
NFT-11	**Security**	Session / token handling	Tampered, expired, forged and alg=none tokens rejected [Target: Project target]	pytest with crafted JWTs; check expiry window	All 401	API: Tampered, expired, wrong-key and alg=none tokens all 401; tokens expire after 120 min; logout revocation covered in FT-41	**Pass**	pytest-results.json
NFT-12	**Security**	Protection of sensitive information	No hashes in responses; passwords/OTPs bcrypt-hashed [Target: PDF NFR06 (partial)]	pytest: scan API responses; inspect stored hashes	No leaks	API: No password/OTP hashes in any API response; 69 passwords bcrypt-hashed; 7 OTPs stored as bcrypt hashes; OTP/QR secret shown only to the receiver (FT-24)	**Pass**	pytest-results.json

TC ID	Quality Attribute	Test Case Description	Metric / Criteria	Test Steps / Tools	Expected Result	Actual Result	Status	Evidence / Defect
NFT-13	**Security**	Brute-force protection	Account locked after 5 failed logins [Target: Project target]	pytest: 6 wrong passwords then correct password	6th attempt -> 429	API: 5 wrong passwords -> 401 each; 6th attempt -> 429; correct password while locked -> 429 'Too many failed attempts. Try again in 15 minutes.'	**Pass**	pytest-results.json
NFT-14	**Security**	File upload validation	Executables, fake PDFs, SVG, empty files rejected [Target: Project target]	pytest: upload malicious / invalid files	415 / 422 / 413	API: Uploads rejected: exe as octet-stream -> 415; html renamed pdf -> 415; empty file -> 422; svg image -> 415; size limit 2 MB -> 413 (FT-35)	**Pass**	pytest-results.json
NFT-15	**Accessibility**	Automated WCAG 2.1 A/AA scan	0 serious or critical violations [Target: Project target]	Playwright + axe-core on 7 pages	No serious issues	UI: 7 pages scanned. /login: 0 serious/critical rule(s); /: 0 serious/critical rule(s); /complaints: 0 serious/critical rule(s); /visitors: 0 serious/critical rule(s); /delivery: 0 serious/critical rule(s); /sos: 0 serious/critical rule(s); /profile: 0 serious/critical rule(s). Total serious/critical nodes: 0	**Pass**	playwright-results.json
NFT-16	**Usability**	Keyboard-only navigation	Login and complaint form usable without a mouse [Target: Project target]	Playwright: Tab / Enter / Escape only	All steps work; skip link first	UI: Login completed with Tab/typing/Enter only; complaint dialog opened with Enter, focus moved into the form, Escape closed it; 'Skip to content' link is first in tab order	**Pass**	DEF-07 (fixed) playwright-results.json
NFT-17	**Usability**	Form validation feedback	Errors shown inline, announced (role=alert), linked to fields [Target: Project target]	Playwright: submit empty registration form	>= 5 linked inline errors	UI: Empty registration submit: 6 inline errors with role=alert; inputs marked aria-invalid and linked via aria-describedby	**Pass**	DEF-07 (fixed) playwright-results.json
NFT-18	**Usability**	Responsive layout	No horizontal scrolling at 375, 768, 1366 px [Target: Project target]	Playwright: 8 pages x 3 widths	No overflow	UI: 8 pages x 3 widths (375, 768, 1366 px) checked; pages with horizontal overflow: none	**Pass**	DEF-02 (fixed) playwright-results.json
NFT-19	**Usability**	New user completes key tasks within 15 minutes	Task time with real first-time users [Target: PDF NFR10]	Moderated usability session with 5 new students	Key tasks < 15 min	Not executed: Needs a moderated session with real first-time users; automated scripts cannot measure human learning time.	**Not Run**	
NFT-20	**Reliability**	Error handling	Correct 404/405/422 JSON errors, no stack traces [Target: Project target]	pytest: missing record, malformed JSON, bad types, unknown route	Clean JSON errors	API: missing record -> 404; malformed JSON -> 422; bad query type -> 422; unknown route -> 404; wrong method -> 405; JSON error bodies, no stack traces; /api/health reports DB status	**Pass**	pytest-results.json
NFT-21	**Data integrity**	Duplicate submissions and race conditions	1 record for 5 parallel identical submits; 1 carrier wins [Target: PDF NFR13]	pytest: 5 parallel complaint POSTs; 4 parallel carrier accepts	No duplicates	API: 5 parallel identical complaint submissions -> codes [201, 409, 409, 409, 409], 1 stored; 4 parallel carrier accepts -> codes [200, 409, 409, 409], carrier recorded once	**Pass**	DEF-01 (fixed) pytest-results.json
NFT-22	**Reliability**	Persistence after server restart	Record readable after restart [Target: PDF NFR13]	Create complaint, restart API process, read it back	Identical data	Complaint CMP-00258 created, API server stopped and restarted, record read back with HTTP 200 and identical data	**Pass**	nft-results.json
NFT-23	**Auditability**	Audit trail and status consistency	All critical actions logged with user + timestamp [Target: PDF NFR12]	pytest: perform approvals, SOS, deliveries; inspect audit_logs	All present	API: Complaint statuses ['CLOSED', 'ESCALATED', 'IN_PROGRESS', 'OPEN', 'RESOLVED'] all valid with matching timestamps; all 10 critical actions present in the audit log with user ID and timestamp	**Pass**	DEF-07 (fixed) pytest-results.json
NFT-24	**Backup & recovery**	Backup and restore	Restore < 2 h with identical row counts [Target: PDF NFR09]	pg_dump of loaded DB, pg_restore into new DB, compare 8 tables	Counts identical	pg_dump 0.8 MB in 0.2 s; restore into a new database in 0.4 s (target < 2 h); row counts of 8 tables identical (e.g. complaints 20,258). Daily scheduling is a deployment task (scripts/backup_db.sh + cron) and was not run here	**Pass**	scripts/backup_db.sh, scripts/restore_db.sh
NFT-25	**Reliability**	Error rate across all load tests	< 0.5 % failed requests [Target: PDF NFR08]	Aggregate of the k6 runs at 50, 150 and 300 users	< 0.5 %	0 failed of 35,598 requests across 3 load runs = 0.000 %	**Pass**	k6-*.json
NFT-26	**Compatibility**	Chromium desktop browser	All E2E tests pass [Target: PDF NFR11]	Playwright Chromium 141 at 1366x850	All pass	Chromium 141.0.7390.37: 21/21 desktop E2E tests passed	**Pass**	playwright-results.json
NFT-27	**Compatibility**	Firefox desktop browser	Smoke test passes [Target: PDF NFR11]	Playwright Firefox project (ALL_BROWSERS=1)	Pass	Not executed – firefox-desktop: Error: browserType.launch: Executable doesn't exist at /opt/pw-browsers/firefox-1495/firefox/firefox	**Blocked**	DEF-06 playwright-results.json
NFT-28	**Compatibility**	Safari / WebKit browser	Smoke test passes [Target: PDF NFR11]	Playwright WebKit project (ALL_BROWSERS=1)	Pass	Not executed – webkit-desktop: Error: browserType.launch: Executable doesn't exist at /opt/pw-browsers/webkit-2215/pw_run.sh	**Blocked**	DEF-06 playwright-results.json

TC ID	Quality Attribute	Test Case Description	Metric / Criteria	Test Steps / Tools	Expected Result	Actual Result	Status	Evidence / Defect
NFT-29	**Compatibility**	Mobile viewports	Login, menu, SOS and forms work on phone screens [Target: PDF NFR11]	Playwright Chromium emulating Pixel 5 and iPhone 12	Pass, no overflow	UI: mobile-pixel5 (393x727, touch): login, hamburger menu, SOS button and complaint dialog work; no horizontal overflow UI: mobile-iphone12-chromium (390x664, touch): login, hamburger menu, SOS button and complaint dialog work; no horizontal overflow	**Pass**	playwright-results.json
NFT-30	**Maintainability**	Linting and type checking	0 lint errors, 0 type errors [Target: Project target]	ESLint, TypeScript strict, Ruff	Clean	ESLint: 0 errors, 0 warnings in 46 files; TypeScript strict: 0 errors; Ruff (Python): 0 issues	**Pass**	eslint.config.js, tsconfig.json, backend/ruff.toml
NFT-31	**Maintainability**	Production build	Build succeeds [Target: Project target]	npm run build (Vite)	Success	Production build succeeded in 8.5 s; JavaScript bundle 341 KB (uncompressed)	**Pass**	frontend/dist
NFT-32	**Portability**	Reproducible setup from scratch	Fresh DB initialises; all env vars documented [Target: Project target]	Create empty DB, run seed, start API, check .env.example	Healthy app	Empty database initialised and seeded in 1.6 s (26 tables, 65 users); API health OK on the fresh DB; .env.example covers 15/15 variables read by the code	**Pass**	backend/app/seed.py, .env.example, Dockerfile
NFT-33	**Availability**	Monthly uptime	>= 99.5 % per month [Target: PDF NFR07]	External uptime monitor on /api/health for 30 days	>= 99.5 %	Not executed: Needs a public deployment monitored for 30 days; the app exposes /api/health for an uptime monitor.	**Not Run**	
NFT-34	**Data privacy**	Personal data encrypted in transit and at rest	PII not readable in database; HTTPS [Target: PDF NFR06]	Inspect stored student/visitor data; check DB TLS	Encrypted	Passwords and OTPs are hashed, but student phone numbers, registration numbers and visitor phone numbers are stored as plain text (e.g. phone '984*******'); local PostgreSQL ssl=off. Encryption in transit relies on HTTPS at the hosting provider (HSTS header is sent in production) and could not be verified locally. NFR06 (encrypted at rest) is NOT met by the application	**Fail**	DEF-04 Defect DEF-04

Non-Functional Testing Summary

Total Test Cases	Executed	Passed	Failed	Blocked	Not Run	Pass Percentage
34	30	28	2	2	2	93.3%

Pass percentage = Passed ÷ Executed × 100. Executed = test cases with a Pass or Fail outcome; Blocked and Not Run are reported separately.

Key Takeaways

- ✔ 30 of 34 non-functional tests were executed: 28 passed, 2 failed (93.3%); 2 blocked and 2 not run.
- ✔ 300 concurrent users: 0.00 % errors, average 1363 ms – only after fixing DEF-03 (database connections exhausted; first run had 0.82 % errors).
- ✔ FAIL NFT-06 – Performance degradation as load increases: see DEF-05.
- ✔ FAIL NFT-34 – Personal data encrypted in transit and at rest: see DEF-04.
- ✔ Security checks passed: 401 on all protected endpoints, 403 role matrix, SQL injection, stored XSS, token tampering, brute-force lockout, upload filtering.
- ✔ Not executed: NFT-19, NFT-27, NFT-28, NFT-33 (other browsers blocked by the sandbox; usability study and 30-day uptime need real users / deployment).
- ✔ SOS alerts reached the warden board in under 5 s even under a 150-user load (NFR03).

5. Performance and load test measurements

Run	Concurrent users	Steady duration	Requests	Throughput (req/s)	Failed %	Avg (ms)	Median (ms)	p95 (ms)	p99 (ms)	Max (ms)
normal-50vus	50	60s	9,966	129.33	0.00	54	56	92	120	261
peak-150vus	150	60s	12,919	167.1	0.00	509	526	781	987	1975
peak-300vus	300	60s	12,713	162.14	0.00	1363	1429	1921	2209	3021
background-sos	150	40s	9,279	162.14	0.00	501	532	813	1037	1410

Tool: k6 v0.54.0. Mixed workload per virtual user: 80 % student session (dashboard, notices, complaints, notifications), 10 % complaint creation, 10 % warden queue reads; 1 s think time; 10 s ramp-up. API: FastAPI/uvicorn with 2 workers, PostgreSQL 16 and the load generator on the same 2-vCPU container, so absolute numbers are lower than dedicated hosting would give. 'background-sos' is the 150-user load used while measuring SOS latency (NFT-04).

6. Requirement traceability

Requirement	Covered by test cases	Result	Requirement	Covered by test cases	Result
FR01	FT-05, FT-06	Pass	**FR26**	FT-34	Pass
FR02	FT-42	Pass	**FR27**	FT-29	Pass
FR03	FT-42	Pass	**FR28**	FT-22, FT-23, FT-27	Pass
FR04	FT-01, FT-02, FT-03, FT-04, FT-31, FT-41	Pass	**FR29**	FT-28	Pass
FR05	FT-07	Pass	**FR30**	FT-24, FT-25, FT-26	Pass
FR06	FT-38, FT-43	Pass	**FR31**	FT-29	Pass
FR07	FT-38, FT-43	Pass	**FR32**	FT-40	Pass
FR08	FT-38	Pass	**FR33**	FT-11, FT-15, FT-30	Pass
FR09	FT-37	Pass	**FR34**	FT-30	Pass
FR10	FT-08, FT-09, FT-11, FT-12	Pass	**FR35**	FT-39	Pass
FR11	FT-08	Pass	**NFR01**	NFT-02, NFT-06	Fail
FR12	FT-13	Pass	**NFR02**	NFT-01, NFT-03, NFT-05	Pass
FR13	FT-14, FT-17	Pass	**NFR03**	FT-20, NFT-04	Pass
FR14	FT-15, FT-16	Pass	**NFR04**	NFT-05, NFT-06	Fail
FR15	FT-18	Pass	**NFR05**	FT-04, FT-12, FT-31, NFT-07, NFT-08, NFT-09, NFT-10, NFT-11, NFT-13, NFT-14	Pass
FR16	FT-19	Pass	**NFR06**	NFT-12, NFT-34	Fail
FR17	FT-03, FT-30	Pass	**NFR07**	NFT-33	Partial
FR18	FT-33	Pass	**NFR08**	FT-26, NFT-20, NFT-25	Pass
FR19	FT-03, FT-39	Pass	**NFR09**	NFT-24	Pass
FR20	FT-43	Pass	**NFR10**	NFT-15, NFT-16, NFT-17, NFT-19	Partial
FR21	FT-36	Pass	**NFR11**	NFT-18, NFT-26, NFT-27, NFT-28, NFT-29	Partial
FR22	FT-20, FT-21	Pass	**NFR12**	NFT-23	Pass
FR23	FT-10	Pass	**NFR13**	FT-27, FT-32, NFT-21, NFT-22	Pass
FR24	FT-35	Pass	**NFR14**	NFT-30, NFT-31, NFT-32	Pass
FR25	FT-35	Pass			

'Partial' = at least one linked case was Blocked or Not Run. FRs from the Lab VI report not covered by an automated case: none.

7. Defect log

Defect ID	Severity	Type	Affected Module	Found by	Reproduction Steps	Expected Result	Actual Result	Status	Resolution
DEF-01	High	Application	Complaints (also visitors, service requests, SOS, deliveries, mess feedback)	NFT-21 (pytest, first run)	Log in as a student; send the same complaint 5 times in parallel (simulated double-click / flaky network).	One complaint stored; the other requests get 409 Conflict.	All 5 requests returned 201 and 5 identical complaints were stored (check-then-insert race).	Fixed	Added a per-user PostgreSQL advisory lock (services.lock_for_user) around duplicate checks in every create endpoint. Retest NFT-21: 1 x 201 + 4 x 409, 1 record stored.
DEF-02	Medium	Application	Visitors (UI layout)	NFT-18 (Playwright, first run)	Log in as a student on a 375 x 812 viewport and open Visitors.	No horizontal scrolling; tables scroll inside their card.	Page width 503 px > 375 px viewport; whole page scrolled horizontally.	Fixed	Cards inside CSS grids now use min-width: 0 so wide tables scroll within the card. Retest NFT-18: no overflow on 8 pages x 3 widths.
DEF-03	High	Application / configuration	Backend database layer	NFT-02 (k6, first run)	Run k6 with 300 virtual users against the API started with 2 workers.	Error rate < 0.5 %, average < 3 s.	0.82 % failed requests, p99 32 s. Log: 'remaining connection slots are reserved' and 'QueuePool limit ... timed out' – 2 workers x 50 pooled connections exceeded PostgreSQL max_connections (100).	Fixed	Pool made configurable (DB_POOL_SIZE / DB_MAX_OVERFLOW / DB_POOL_TIMEOUT) with defaults sized to the 40-thread request pool (35 + 5 per worker). An intermediate setting (10 + 5) caused 7 % pool time-outs and was rejected. Retest at 300 users: 0 % errors, average 1.4 s.
DEF-04	Medium	Application (requirement gap)	Data storage	NFT-34	Inspect the users and visitor_requests tables in PostgreSQL.	Personal data (phone, registration number, visitor details) encrypted in storage.	Passwords and OTPs are bcrypt-hashed, but phone numbers, registration numbers and visitor phone numbers are plain text.	Open	Planned: column-level encryption (pgcrypto or application-side Fernet) for PII, plus a managed database with encryption at rest and TLS (sslmode=require).
DEF-05	Medium	Capacity	Whole system (single 2-vCPU host)	NFT-06 (k6)	Run k6 at 50, 150 and 300 users on the same 2-vCPU machine that also runs PostgreSQL and the load generator.	No significant degradation (p95 at 300 users <= 2 x p95 at 50 users).	p95 92 ms at 50 users vs 1,921 ms at 300 users; throughput levels off at about 162 requests/s (CPU-bound). Still within the 3 s average target.	Open	Re-test on production-sized hosting; scale horizontally (more workers/instances), cache dashboard counts, move the load generator off the server.
DEF-06	Low	Test environment	Compatibility testing	NFT-27, NFT-28	Run 'ALL_BROWSERS=1 npx playwright test'.	Smoke test runs in Firefox and WebKit.	browserType.launch: Executable doesn't exist – the sandbox network policy blocks the Playwright browser downloads.	Open (blocked)	On a normal machine run 'npx playwright install firefox webkit' then 'npm run test:all-browsers' in e2e/.
DEF-07	Low	Test script	Automated tests	First runs: NFT-23 (pytest); FT-06, FT-08, FT-30, NFT-10, NFT-16, NFT-17 (Playwright)	(a) NFT-23 checked audit entries created by tests that run later; (b) E2E locators such as getByLabel('Category') also matched filter/search inputs; (c) a CSS selector used a React id containing ':'; (d) a Tab key press happened before the page rendered; (e) run_all_tests.sh stored the PID of a wrapper shell instead of the load-test server, so the server kept running and later exhausted PostgreSQL connections (HTTP 500 in the NFT runner).	Tests independent of order and with unambiguous selectors.	7 false failures in the first runs (1 pytest, 6 Playwright), 3 more while fixing selectors, and one aborted pipeline run (e) – none were application defects.	Fixed	NFT-23 now performs its own actions; E2E tests use name-based selectors and wait for rendering; the pipeline starts the server with 'exec' and waits for the port to close. All affected tests were re-run with the complete pipeline.
DEF-08	Low	Application	UI components (Visitors, Services, SOS, Gate desk)	Manual visual review of screenshots (not detected by automated tests)	Log in as a warden and open Visitors.	Card title on the left, compact filter on the right.	The filter used the full-width input style, so it filled the header and wrapped the card title onto two lines.	Fixed	FilterSelect now has its own auto-width style. Frontend unit tests and the full Playwright suite were re-run afterwards.
DEF-09	High	Application	Authentication / data layer	User report – local run on Windows with SQLite (not covered: automated suites used PostgreSQL)	Start the app without DATABASE_URL (SQLite file), log in once, then log in again.	Login succeeds.	TypeError: can't compare offset-naive and offset-aware datetimes in auth.login – SQLite returns dates without a time zone.	Fixed	All date/time columns now use a UTCDateTime type that always returns time-zone-aware UTC values. Re-tested: backend suite 60/61 and browser suite 23/23 on SQLite (the remaining SQLite case is the PostgreSQL-only parallel-submit lock, NFT-21); 61/61 and 23/23 on PostgreSQL.

8. Test environment and limitations

Linux 6.18.44-fc-v139 · 2 vCPU · Python 3.13.16 · Node v22.22.0 · psql (PostgreSQL) 16.15 (Ubuntu 16.15-0ubuntu0.24.04.1) · k6 v0.54.0 (commit/baba871c8a, go1.23.1, linux/amd64) · Chromium 141.0.7390.37. All tests executed locally in a cloud Linux container (2 vCPU). The API, PostgreSQL and the load generator share the same machine.

- The application was not deployed to a public URL from this environment (no hosting account access). All results come from a local run; deployment files (Dockerfile, render.yaml) and instructions are included.
- Load tests ran on one 2-vCPU container that also hosted PostgreSQL and the k6 load generator; production hosting will give different numbers.
- Firefox and WebKit/Safari could not be downloaded in this sandbox (network policy), so cross-browser results are limited to Chromium and mobile-viewport emulation in Chromium. Real Android/iOS devices were not tested.
- NFR10 (key tasks within 15 minutes) and NFR07 (99.5 % monthly uptime) need real users and a long-running deployment and were not executed.
- Payments are simulated demo records (no payment gateway). SOS alerts are in-app only and are not connected to emergency services.
- Encryption at rest of personal data (NFR06) is not implemented (DEF-04).